

Ahmed Ayyan Mukhtar

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Education

Information Technology University, Lahore

2023 – 2027

B.Sc. Software Engineering

Work Experience

Front-End Developer Intern — *CyberSoft Solutions, Lahore*

Feb – Jul 2025

- Shipped production UI for the UBL banking app (Angular 17); built a reusable component library cutting per-feature dev time ~25%; applied lazy loading + OnPush change detection to meet banking-grade SLAs.

Tech Lead — “Code Rush” — *GDG on Campus, ITU*

2024 – Present

- Organised university-wide competitive programming contest (100+ participants): problem design, platform setup, live troubleshooting.

Projects

MediSense AI — Hybrid RAG Clinical Assistant

Python, LangChain, ChromaDB

- Architected a hybrid RAG pipeline over WHO/NIH medical guidelines and OCR-scanned patient reports; multi-collection ChromaDB design maintains isolated vector spaces per user for context-aware, patient-specific retrieval.
- Sliding-window chunking (500-char chunks, 50-char overlap) with all-MiniLM-L6-v2 embeddings; dual retrieval (top-5 global + top-3 personal) fed into a LangChain + PydanticOutputParser chain enforcing strict JSON schema to prevent hallucinated output in a safety-critical domain.
- 5th / 200+ teams — HEC Generative AI National Hackathon 2025.

Reply Mirror — Multi-Agent Fraud Detection Pipeline

LangGraph, Langfuse, Python

- Credit-aware 7-agent pipeline (Ingest → Behavior → Geo → Comms → Audio → Fusion → Decision) routing LLM calls only to high-risk users; deterministic anomaly scoring fused with LLM phishing/vishing scores via a weighted multiplicative model with guard rails.
- Otsu-style adaptive thresholding for final fraud labelling; full per-run observability across all agent stages via Langfuse — built specifically to test and monitor agent performance over time.

CI/CD Build Failure Predictor

XGBoost, SHAP, FastAPI, Docker, AWS

- Trained and compared Logistic Regression, Random Forest, and XGBoost on 20,000 parsed Travis CI build logs, engineering commit-metadata and log-derived features (error count, exit code, prior failure rate); Optuna-tuned XGBoost selected as production model (F1 0.49, AUC-ROC 0.77) with decision threshold tuned to prioritize catching failed builds.
- Used SHAP to surface the strongest failure predictors (exit code, error count, prior failure rate, log line count, files changed) for interpretable pre-build risk signals.
- Exposed the model as a REST API via FastAPI to serve as a pre-build gate for real CI pipelines; containerized with Docker and deployed to AWS Elastic Beanstalk with an interactive UI for testing preset logs, custom log uploads, and manual feature entry.

A-EYE J.A.R.V.I.S. — Local Multimodal AI Assistant

Ollama, YOLOv8, FastAPI, Angular

- Fully local multimodal assistant combining YOLOv8, Ollama LLMs, faster-whisper, Silero VAD, and MediaPipe; INT8-quantized models with FastAPI + Angular front end over WebSockets, zero cloud dependency. Capability-aware multi-model routing across chat, code, and vision roles with JSON-schema tool calling and an auto-detected fallback protocol for models without native tool support.

SentinelAI — AI Code Review VS Code Extension

VS Code API, Groq, Next.js

- Dual-component tool: VS Code extension (Diagnostics, Hover, Code Actions API) + Next.js microservice running three parallel LLM agents (security, complexity, code smell) via Groq and OpenRouter; SHA-256 caching for ~0ms cache hits, debounced auto-analysis, one-click Quick Fixes across 11 languages.

Technical Skills

Languages: Python, TypeScript, JavaScript, C++, PHP, HTML5/CSS3/SCSS

AI / ML: LangChain, LangGraph, RAG pipelines, XGBoost, SHAP, scikit-learn, Optuna, YOLOv8, MediaPipe, faster-whisper, Ollama, Groq, OpenRouter, Gemini API, Langfuse, all-MiniLM-L6-v2

Cloud / DevOps: AWS (Elastic Beanstalk), Docker, Git, GitHub Actions, Vercel

Backend / Data: FastAPI, Node.js, Express, PostgreSQL, MySQL, MongoDB, SQLite, ChromaDB, Supabase (RLS · Real-time)

Frameworks: React 19, Next.js 14/15, Angular 17, Laravel 11

Tooling: Figma, Jira, TanStack Query, Zod, PyQt6, VS Code Extension API

Achievements

- 5th / 200+ teams — HEC Generative AI National Hackathon 2025 (MediSense AI). Delivered live client sites (Orionex, Ideal Janitorial Services, Human Rights Organisation) — SEO-optimised, production-hosted; 20+ public projects on GitHub.